

company specific factor loadings. In the explicit approaches analysts begin with either a set of specified factors and estimate sensitivities to those factors (i.e., index models and macroeconomic factor model) or begin with the factor loadings (fundamental data) and estimate the set of explanatory factors (cross-sectional model).

The advantage of a statistical factor model over the previously described explicit approaches is that it provides risk managers with a process to uncover accurate covariance and correlation relationships of returns without making any assumptions regarding what is driving the returns. Any preconceived bias is removed from the model. The disadvantage of this statistical approach is that it does not provide portfolio managers with a set of factors to easily determine what is driving returns since the statistical factors do not have any real-world meaning.

To the extent that analysts are only interested in uncovering covariance and correlation relationships for risk management purposes, PCA has proven to be a viable alternative to the traditional explicit modeling approaches. Additionally, with the recent growth of exchange traded funds (ETFs) many managers have begun correlating their statistical factors to these ETFs in much the same way Ross and Roll did with economic data to better understand these statistical factors.

The process to derive the statistical model is as follows:

Step 1. Compute the sample covariance matrix by definition from historical data. This matrix will likely suffer from spurious relationships due to the data limitations (not enough degrees of freedom and potential false relationships). But these will be resolved via principal component analysis.

Let $\bar{C}$ represent the sample covariance matrix.

Step 2. Factor the sample covariance matrix. We based the factorization scheme on eigenvalue-eigenvector decomposition. This is:

$$\bar{C} = VD{V'} \quad (6.42)$$

where D is the diagonal matrix of eigenvalues sorted from largest to smallest, $\lambda_1 > \lambda_2 > \dots > \lambda_n$ and V is the corresponding matrix of eigenvectors and these eigenvalues are determined by computing the percentage of total variance that is explained by each eigenvector. In finance, the terminology that is most often used when referring to determining the eigenvalue with the